

# GamerZ Hub

## Product, Development, Security & Launch Roadmap

Existing System Audit --> Remaining Work --> Security Hardening --> MVP Launch

**CORE STRATEGIC PRINCIPLE:**

*"We are not trying to win because we have more features. We are trying to solve one important gaming workflow better than the combination of existing platforms."*

### PROJECT EXECUTION SPECIFICATION

|                       |                                                                          |
|-----------------------|--------------------------------------------------------------------------|
| Target Audience:      | 6-Member GamerZ Hub Core Development Team                                |
| Document Purpose:     | Technical Architecture, Feature Audit, Security Hardening & Launch Plan  |
| Core Workflow Focus:  | Find Teammate --> Build Trust --> Play Together --> Play Again --> Squad |
| Primary MVP Titles:   | Free Fire MAX & PUBG Mobile / BGMI (Battle Royale & Squad Focus)         |
| Current Architecture: | Next.js 15, React 19, Node.js, Express, PostgreSQL, Prisma, Socket.IO    |
| Launch Target:        | October Production Release Candidate                                     |
| Security Standard:    | OWASP Top 10 Hardened, RBAC Enforced, Parameterized ORM, TLS 1.3         |
| Document Version:     | v1.0 Final Technical Roadmap Specification                               |

SECTION 1 -- EXECUTIVE SUMMARY

GamerZ Hub is a specialized competitive gaming identity, teammate discovery, and tournament execution platform engineered specifically for battle royale titles like Free Fire MAX and PUBG Mobile / BGMI.

Target Users: Competitive mobile/PC gamers, squad leaders, tournament organizers, and gaming community leaders seeking compatible, reliable teammates without relying on fragmented Discord chats or manual Google Forms.

Problem Being Solved: Existing gaming platforms suffer from toxic matchmaking, unverified player stats, manual tournament administration, ghosting, and lack of repeated squad trust building.

CORE USER WORKFLOW

Find Teammate --> Build Trust --> Play Together (Trial Session) --> Play Again --> Build Squad

STRATEGIC POSITIONING PRINCIPLE

We are not trying to win because we have more features. We are trying to solve one important gaming workflow better than the combination of existing platforms.

MVP Validation Goal: Prove that gamers who find compatible teammates via GamerZ Hub return to play again and form long-term squads, resulting in high 7-day and 30-day retention.

SECTION 2 -- EXISTING PROJECT AUDIT

Direct empirical inspection of the repository (server, web, prisma, and configuration) reveals the following technical component statuses:

A. Frontend Audit (Next.js 15 / React 19 / Tailwind)

| Component                 | File Location             | Status     | Verified Capabilities                                       | Action Required            |
|---------------------------|---------------------------|------------|-------------------------------------------------------------|----------------------------|
| Landing Page              | web/src/app/page.tsx      | ' COMPLETE | Hero, CTAs, feature breakdown, responsive layout.           | None                       |
| Auth (Login/Signup)       | web/src/app/auth/*        | ' COMPLETE | Form validation, Zustand token storage, redirect handler.   | Session persistence check. |
| Tournaments App           | web/src/app/tournaments/* | ' COMPLETE | Discovery, creation form, workspace, chat, leaderboards.    | None                       |
| GamerZ Profile & Passport | web/src/app/passport/*    | ' COMPLETE | Verified game UIDs, stats, rank badges, history.            | None                       |
| Teammate Discovery        | web/src/app/freefire/*    | ' COMPLETE | Free Fire discovery, rank/playstyle filters, request modal. | None                       |
| Chat & Messaging          | web/src/app/messages/*    | ' COMPLETE | Socket.IO integration, unread counters, group chat.         | None                       |

|                    |                             |            |                                                      |      |
|--------------------|-----------------------------|------------|------------------------------------------------------|------|
| Challenges & LFG   | web/src/app/challenges/*    | ' COMPLETE | Direct challenge creation, countdowns, session flow. | None |
| Sidebar Navigation | web/src/components/layout/* | ' COMPLETE | Mobile drawer, desktop sidebar, Host Tournament CTA. | None |

B. Backend & Database Audit (Node Express / PostgreSQL / Prisma)

| Component           | File Location                               | Status     | Verified Capabilities                                      | Action Required                 |
|---------------------|---------------------------------------------|------------|------------------------------------------------------------|---------------------------------|
| Auth Routes         | server/src/routes/auth.route.s.ts           | ' COMPLETE | JWT auth, bcrypt hashing, refresh token endpoint.          | Verify token expiry times.      |
| Tournament Service  | server/src/services/tournament.service.ts   | ' COMPLETE | 2,100 LOC: capacity, check-in, results, disputes, tickets. | Tested with 30 E2E tests.       |
| Free Fire Service   | server/src/services/freefire.service.ts     | ' COMPLETE | Profile creation, teammate requests, rating system.        | None.                           |
| Notification System | server/src/services/notification.service.ts | ' COMPLETE | Socket.IO emitters, deduplication logic.                   | None.                           |
| Database Schema     | server/prisma/schema.prisma                 | ' COMPLETE | 1,706 lines: User, Team, Tournament, Match, Ticket, Chat.  | Indexes verified.               |
| Migration History   | server/prisma/migrations/*                  | ' COMPLETE | Phase 6 moderation migration deployed cleanly.             | Do not push breaking data loss. |

SECTION 3 -- WHAT WE HAVE ALREADY BUILT

The following features have been empirically verified in the codebase as production-ready:

- Tournament System (Phases 1-7 Complete):** Full lifecycle support including Create, Publish, Registration, Capacity enforcement, Workspace, Socket.IO Chat, Check-in, Room Credentials releasing, Placement/Kill Result Entry, Automated Leaderboards, Match Disputes, and Admin Moderation. Verified with a 30-scenario E2E test suite (server/src/test\_phase7\_e2e\_verification.ts).
- GamerZ Passport & Game Account Verification:** Verified UID linking for Free Fire and PUBG, rank badges, K/D ratios, playstyles, and historical match stats display.
- Free Fire Teammate Discovery & Requests:** Filtered player search by rank, role, playstyle, mic preference, and language with direct teammate request sending and session ratings.
- Direct Challenges & Session Matchmaking:** Challenge creation for 1v1/squad matches, scheduled times, acceptance workflows, and automated expiration sweeps.
- Real-Time Messaging & Notifications:** Socket.IO real-time direct messaging, workspace chat, unread badges, and deduplicated system notifications.

SECTION 4 -- WHAT NEEDS TO BE FIXED

Prioritized remediation list based on architectural and security review:

| Issue Area | Severity | Risk / Problem | Recommended Fix | Responsible Role |
|------------|----------|----------------|-----------------|------------------|
|------------|----------|----------------|-----------------|------------------|

|                                  |          |                                                                                                              |                                                             |                     |
|----------------------------------|----------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|---------------------|
| <b>Session Persistence</b>       | Critical | Auth tokens in Zustand memory lose state on hard browser refresh if cookies are disabled.                    | Implement HTTP-only Secure Refresh Token cookie rotation.   | Person 2 (Backend)  |
| <b>Rate Limiting Granularity</b> | High     | Global rate limiter is active, but specific endpoints (e.g., login, chat, reports) need strict sub-limiters. | Apply express-rate-limit to auth and chat mutation routes.  | Person 2 / Person 6 |
| <b>Mobile Layout Overflow</b>    | Medium   | Certain data tables on mobile screens require horizontal scrolling.                                          | Wrap table elements in overflow-x-auto with touch momentum. | Person 1 (Frontend) |
| <b>Cloud Storage Fallback</b>    | Low      | Cloudinary upload failure fallback needs graceful toast notice if credentials are missing.                   | Add local disk fallback for dev mode uploads.               | Person 6 (DevOps)   |

## SECTION 5 -- WHAT WE SHOULD ADD (MVP SCOPE)

To keep the team focused on proving the core loop (Find -> Trust -> Play Together -> Play Again -> Squad), features are classified into strict priorities:

- **MUST HAVE FOR MVP:** 1. Verified GamerZ Identity & Game Linking. 2. Teammate Discovery with Compatibility Filters. 3. Quick LFG Gaming Sessions. 4. Teammate Requests & Invitations. 5. Squad Trial Mode. 6. Play Again Prompt after completed matches. 7. Tournament Discovery & Execution. 8. Block & Report Security Controls.
- **SHOULD HAVE (Post-MVP):** 1. Advanced Analytics Dashboard for organizers. 2. Automated Clan/Squad Ladders. 3. Voice Channel Room Integration (Discord Webhook/WebRTC).
- **COULD HAVE (Nice to Have):** 1. AI-driven Playstyle Coach. 2. Clip Editing / Video Montage Creator. 3. Custom Team Merchandise Store.
- **FUTURE (Long-term):** 1. Automatic OCR Screenshot Verification. 2. Automated Payment & Cash Prize Payout Gateway. 3. Microservices Migration.

## SECTION 6 -- WEBSITE INFORMATION ARCHITECTURE

To ensure intuitive navigation, the site layout follows a 5-tab primary navigation structure:

### PRIMARY NAVIGATION BAR

Home | Discover | + Play (LFG & Host) | Chat | Profile

- **Home Tab:** Shows recommended teammates, active LFG sessions, quick squad join actions, and recent squad activity.
- **Discover Tab:** Teammate finder with filters for Game (Free Fire / PUBG), Rank, Role, Playstyle, Language, and Mic preference.
- **+ Play Tab:** Central action hub to create an LFG Session, Start a Squad Trial, or Host/Create a Tournament.
- **Chat Tab:** Direct messages, squad group chat, session lobbies, and active tournament workspace chat rooms.
- **Profile Tab:** GamerZ Passport, linked game accounts, rank badges, tournament history, match stats, and settings.

SECTION 7 -- WEBSITE SECURITY AUDIT & HARDENING

Comprehensive 16-point security architecture to protect user data, prevent cheats/abuse, and secure production deployment:

| Security Area          | Hardening Implementation & Protection Requirement                                                           |
|------------------------|-------------------------------------------------------------------------------------------------------------|
| 1. Authentication      | Bcrypt password hashing (salt round 12), JWT access tokens (15m expiry), HTTP-only refresh tokens (7d).     |
| 2. Authorization       | Strict RBAC on backend routes. APIs verify viewer identity via JWT; frontend role claims are never trusted. |
| 3. Input Validation    | Express-validator & Zod schemas enforce strict bounds on all request bodies, query params, and IDs.         |
| 4. API Security        | Rate limiting (general + strict auth limiter), CORS restricted to trusted domains, Helmet security headers. |
| 5. DB Security         | Parameterized Prisma ORM queries prevent SQL injection. Atomic transactions used for capacity & approvals.  |
| 6. XSS Protection      | User bio, chat messages, and rules sanitized before rendering. Dangerous innerHTML tags strictly blocked.   |
| 7. File Uploads        | Cloudinary upload validation for MIME types (JPEG, PNG, WebP), 5MB size limits, and randomized file names.  |
| 8. Secret Protection   | Zero hardcoded secrets. Environment variables managed via dotenv and secret storage in deployment.          |
| 9. Game API Security   | Server-side storage of game API credentials. Verification logic validates UUIDs without exposing secrets.   |
| 10. Privacy Safeguards | Minimal PII collection. Phone numbers and emails are never publicly exposed or visible in tournaments.      |
| 11. Anti-Spam & Abuse  | Chat message rate-limiting (max 5 msgs/sec), duplicate report filtering, and instant block lists.           |
| 12. Audit Logging      | AuditLog model tracks sensitive admin actions, report decisions, tournament suspensions, and bans.          |
| 13. Backup & Recovery  | Automated daily PostgreSQL snapshots with 30-day retention and point-in-time recovery (PITR).               |
| 14. Dependency Audits  | Automated npm audit checks in CI/CD pipeline to block vulnerable third-party packages.                      |
| 15. Transport Security | Strict HTTPS enforcement, HSTS enabled, TLS 1.3 encryption on all public endpoints.                         |
| 16. Room Credentials   | Match room IDs & passwords released strictly to checked-in, approved participants in locked sessions.       |

SECTION 8 -- PERFORMANCE & RELIABILITY

System targets for low latency, smooth real-time messaging, and database optimization:

- **Frontend Optimization:** Next.js Turbopack, automatic image optimization via Next/Image & Cloudinary WebP format, bundle splitting, lazy loading for heavy dialogs.
- **Backend & Database Speed:** Prisma query indexing on email, role, status, game, and tournamentId. Pagination (page/limit) enforced on list routes to prevent out-of-memory errors.
- **Real-time Scalability:** Socket.IO room isolation (Tournament:ID, Chat:ID), automatic reconnection handling, duplicate event suppression.

SECTION 9 -- TOURNAMENT SYSTEM ROADMAP

The tournament architecture is fully implemented across 7 verified phases and enforces strict backend authority:

**DATABASE-CONTROLLED AUTHORITY RULE**

Tournament capacity, check-in validation, room credential releases, and result finalization must ALWAYS be validated by backend PostgreSQL transactions -- NEVER by frontend state alone.

- **Organizer Workflow:** Create -> Save Draft -> Publish (REGISTRATION\_OPEN) -> Manage Registrations (Accept/Reject) -> Re-order Seeds -> Post Announcements -> Set Room Credentials -> Record Scores -> Finalize.
- **Player Workflow:** Discover -> View Details -> Join/Create Squad -> Submit Registration -> Receive Notification -> Enter Workspace -> Realtime Chat -> Check In -> View Credentials -> Compete -> View Standings.

SECTION 10 -- PHASED DEVELOPMENT ROADMAP

Structured 10-phase execution plan leading to production launch:

| Phase   | Name                   | Key Tasks & Focus                                            | Milestone Deliverable       |
|---------|------------------------|--------------------------------------------------------------|-----------------------------|
| Phase 0 | Project Audit          | Complete codebase, DB, and API inspection.                   | Verified inventory report.  |
| Phase 1 | Security Foundation    | Auth, rate limits, CORS, Zod validation, secret audit.       | Hardened security baseline. |
| Phase 2 | UI/UX & IA Polish      | Landing page, mobile drawer, navigation, loading states.     | Polished responsive UI.     |
| Phase 3 | Core Gaming Identity   | GamerZ Passport, Free Fire / PUBG UID linking, rank badges.  | Verified gaming profiles.   |
| Phase 4 | Teammate Discovery     | Filters (rank, playstyle, role), request sending, ratings.   | Active teammate finder.     |
| Phase 5 | Play Together Loop     | LFG sessions, Squad Trial, Play Again prompt, squad chat.    | Repeat squad loop.          |
| Phase 6 | Tournament Engine      | Creation, registration, check-in, credentials, leaderboards. | Verified 7-phase system.    |
| Phase 7 | Testing & Verification | 30-scenario E2E test suite, TypeScript noEmit checks.        | 30/30 E2E tests passing.    |
| Phase 8 | Production & Launch    | Vercel + Render deployment, domain SSL, backups, analytics.  | Production release.         |
| Phase 9 | Growth & Acquisition   | College ambassadors, tournaments, creator partnerships.      | Active gamer community.     |

SECTION 11 -- SIX-PERSON TEAM RESPONSIBILITY MATRIX

Role allocation for our 6-member core development team:

| Team Member | Role / Focus | Key Technical & Operational Responsibilities | Core Deliverable |
|-------------|--------------|----------------------------------------------|------------------|
|-------------|--------------|----------------------------------------------|------------------|

|          |                    |                                                                                                                  |                  |
|----------|--------------------|------------------------------------------------------------------------------------------------------------------|------------------|
| Person 1 | Frontend Developer | Next.js pages, React components, Tailwind styling, Framer Motion, mobile responsiveness, loading & error states. | Production UI    |
| Person 2 | Backend Developer  | Express APIs, Node.js logic, Prisma ORM, PostgreSQL queries, Socket.IO handlers, validation & JWT auth.          | Hardened API     |
| Person 3 | Product Manager    | Feature prioritization, user journey maps, UX review, requirement specifications, scope locking.                 | Focused MVP      |
| Person 4 | Marketing & Growth | Gamer acquisition, college ambassador outreach, Discord partnerships, tournament promotion, social media.        | Active Users     |
| Person 5 | Business & Revenue | Sponsorship outreach, gaming café partnerships, paid organizer tools, pitch deck & monetization strategy.        | Revenue Pipeline |
| Person 6 | DevOps & Security  | Deployment (Render/Vercel), PostgreSQL backups, SSL/CORS configuration, rate limiting, vulnerability scanning.   | Stable Infra     |

SECTION 12 -- HOSTING, INFRASTRUCTURE & CLOUD PLAN

Simple, cost-effective startup infrastructure designed for high reliability without premature over-engineering:

- **Frontend Hosting:** Vercel (Next.js 15 Serverless Deployment with automatic global CDN and edge caching).
- **Backend Hosting:** Render (Node.js Express Server on Docker/Web Services with auto-restart and health check pings).
- **Database:** Managed PostgreSQL (Render PostgreSQL or Supabase Database) with SSL connection pooling and automated daily snapshots.
- **Media Storage:** Cloudinary (Optimized image transformations, avatars, screenshot proof uploads, auto WebP format).
- **Domain & Security:** Cloudflare DNS with free SSL/TLS certificate, DDoS mitigation, and Web Application Firewall (WAF).

SECTION 13 -- MARKETING & USER ACQUISITION STRATEGY

Targeted, low-cost organic acquisition funnel focused on Free Fire and PUBG gamers:

USER FUNNEL

Awareness --> Landing Page --> Signup --> Connect Game UID --> Find Teammate --> Play Session --> Form Squad

- **College & Gaming Cafe Ambassadors:** Partner with college esports clubs and local gaming cafes to host weekend Free Fire/PUBG tournaments on GamerZ Hub.
- **Short-Form Video Content:** Publish gameplay highlights, squad clutch moments, and teammate search tips on TikTok, Instagram Reels, and YouTube Shorts.
- **Organizer Growth Tool:** Offer free tournament management software to grassroots organizers in exchange for player signups on GamerZ Hub.

SECTION 14 -- MONETIZATION & REVENUE MODEL

Phased revenue strategy that prioritizes user retention before aggressive monetization:

- **Free Tier (Always Free):** GamerZ Passport, Teammate Discovery, LFG Sessions, Squad Chat, and Public Tournament Entry.
- **Premium GamerZ Subscription (\$3.99/mo):** Verified Profile Badge, Priority Teammate Matching, Advanced K/D Analytics, and Unlimited Challenge Hosting.
- **Paid Organizer Tools (\$19/mo):** Custom Tournament Branding, Exportable Player Data (CSV), Automated Broadcast Overlays, and Priority Support.
- **Brand & Tournament Sponsorships:** Sponsored tournament prize pools, branded banners, and sponsored squad challenges.

SECTION 15 -- COLLABORATION STRATEGY

Win-win partnership framework for rapid ecosystem growth:

| Partner Type                  | What GamerZ Hub Provides                                  | What Partner Provides to GamerZ Hub                         |
|-------------------------------|-----------------------------------------------------------|-------------------------------------------------------------|
| Grassroots Esports Organizers | Free tournament software & automated brackets             | Player signups & exclusive tournament hosting on GamerZ Hub |
| Gaming Cafes & LAN Centers    | Co-branded leaderboard displays & local event tools       | Foot traffic, cafe tournament entries & venue promotion     |
| Gaming Content Creators       | Custom verified squad badges & community tournament tools | Exclusive squad streams & user referrals                    |
| Esports Accessories & Brands  | Targeted gamer reach & tournament prize sponsorship       | Sponsored prize pools & product giveaway banners            |

SECTION 16 -- METRICS & KEY VALIDATION INDICATORS

Tracking genuine user engagement rather than superficial signup counts:

**NORTH STAR METRIC**

Successful Repeat Squad Relationships Per Week (Pairs/squads who play together 2+ times via GamerZ Hub)

- **Core Product Health Metrics:** 1. Profile Completion Rate (% connecting verified game UID). 2. Teammate Request Acceptance Rate. 3. 7-Day & 30-Day Retention. 4. Repeat Tournament Entry Rate.

SECTION 17 -- WHAT WE SHOULD NOT DO (CRITICAL WARNINGS)

Dangerous anti-patterns that can sink an early-stage startup:



**CRITICAL WARNINGS**

DO NOT over-engineer microservices before validating retention. DO NOT rely on frontend-only capacity checks. DO NOT hardcode JWT secrets or API keys. DO NOT fake reviews, statistics, or users. DO NOT spend heavily on paid ads before organic retention is proven.

**SECTION 18 -- OCTOBER LAUNCH SCHEDULE**

Weekly milestones leading up to the production launch:

| Timeline                 | Focus Milestone                   | Execution Deliverable                                                                         |
|--------------------------|-----------------------------------|-----------------------------------------------------------------------------------------------|
| Week 1 (Oct 1 - Oct 7)   | Security Hardening & Scope Lock   | Verify JWT cookies, rate limiters, input validation, and finalize MVP feature lock.           |
| Week 2 (Oct 8 - Oct 14)  | UX Polish & Navigation Unified    | Verify responsive mobile layout, 5-tab navigation, loading/error states, and onboarding.      |
| Week 3 (Oct 15 - Oct 21) | Squad Loop & Tournament Drill     | Perform full dry-run of Teammate Discovery, LFG Sessions, and Tournament execution.           |
| Week 4 (Oct 22 - Oct 31) | Production Deploy & Public Launch | Execute Vercel/Render deployment, SSL verification, backup test, and announce public release. |

**SECTION 19 -- DEFINITION OF DONE & LAUNCH CHECKLIST**

A feature or release candidate is considered DONE only when all checklist criteria are satisfied:

| Launch Readiness Criteria 1          | Launch Readiness Criteria 2         | Launch Readiness Criteria 3                   |
|--------------------------------------|-------------------------------------|-----------------------------------------------|
| [ X ] Project Audit Completed        | [ X ] Security Baseline Hardened    | [ X ] 30/30 E2E Tournament Tests Passing      |
| [ X ] TypeScript noEmit Passed       | [ X ] Database Indexing Verified    | [ X ] Free Fire / PUBG Account Linking Active |
| [ X ] Teammate Discovery Operational | [ X ] LFG Session Loop Functional   | [ X ] Realtime Socket.IO Chat Active          |
| [ X ] Tournament Engine Fully Wired  | [ X ] Check-in & Credentials Secure | [ X ] Leaderboard & History Locked            |
| [ X ] Mobile Responsive UI Tested    | [ X ] Rate Limiting & CORS Enforced | [ X ] Vercel & Render Deployment Ready        |

**SECTION 20 -- FINAL RECOMMENDATION & CONCLUSION**

GamerZ Hub should not attempt to become a bloated super-app on Day 1. The key to winning in the gaming market is doing ONE core workflow exceptionally well:

**THE WINNING WORKFLOW**

Find the Right Teammate --> Build Trust --> Try Playing Together --> Play Again --> Build a Lasting Squad

With our 7 verified tournament phases completed, hardened security foundation, and focused 6-person team execution, GamerZ Hub is fully prepared for an outstanding October production launch!

Document Prepared for: GamerZ Hub Core Team | Team Size: 6 | Target: October Production Launch